

Window bounds for repetitions and block complexity of algebraic numbers

Abstract

Let ξ be a real algebraic irrational number and $b \geq 2$ an integer. Write $R(L)$ for the length of the shortest prefix of its fractional base- b expansion containing two occurrences of some word of length L , and $p(L)$ for its block complexity. For every fixed $\theta > 1$ there is $c > 0$ such that every sufficiently large interval $[X, X^\theta]$ contains an integer L for which

$$R(L) > cL\sqrt{\frac{\log L}{\log \log L}}, \quad p(L) > \frac{c}{2}L\sqrt{\frac{\log L}{\log \log L}}.$$

Consequently the logarithmically weighted complexity limsup is infinite for every weight exponent $\gamma > 1/2$. We also obtain windows $[X, X^{1+o(1)}]$ below this endpoint. The arithmetic input is a finite-support moving-weight sparse-parameter theorem proved here by adapting the auxiliary-polynomial argument of Evertse and Ferretti. The additional combinatorial step retains all dyadic prefix scales: when several scales yield the same approximation scale, their multiplicity gives an exponential improvement in approximation quality. Summing fixed-quality counting bounds removes a logarithmic loss without requiring an extension with a different smallness exponent at each point.

1 Introduction

Fix a real algebraic irrational number ξ and an integer $b \geq 2$. We may assume $0 < \xi < 1$, and write $\xi = \sum_{i \geq 1} d_i b^{-i}$, with $d_i \in \{0, \dots, b-1\}$. Define

$$p(L) = \#\{d_{j+1} \cdots d_{j+L} : j \geq 0\}.$$

Following Bugeaud and Kim [4], let $R(L)$ be the least prefix length containing two, possibly overlapping, occurrences of a word of length L . We use R rather than r to distinguish this function from the preperiod length of a rational approximation. Pigeonholing gives

$$R(L) \leq L + p(L). \tag{1.1}$$

Indeed, the prefix of length $L + p(L)$ contains $p(L) + 1$ starting positions for length- L words. All logarithms without a subscript are natural, and iterated logarithms are used only for sufficiently large arguments.

Theorem 1.1. *For every fixed $\theta > 1$, there is a constant $c = c(\xi, b, \theta) > 0$ such that, for every sufficiently large real X , there is an integer $L \in [X, X^\theta]$ satisfying*

$$R(L) > cL\sqrt{\frac{\log L}{\log \log L}}, \quad p(L) > \frac{c}{2}L\sqrt{\frac{\log L}{\log \log L}}. \tag{1.2}$$

Corollary 1.2. *For each $g \in \{R, p\}$,*

$$\limsup_{L \rightarrow \infty} \frac{g(L) \sqrt{\log \log L}}{L \sqrt{\log L}} > 0.$$

For every $\gamma > 1/2$ and every $\eta < 1/2$,

$$\limsup_{L \rightarrow \infty} \frac{g(L) (\log \log L)^\gamma}{L \sqrt{\log L}} = \infty, \quad \limsup_{L \rightarrow \infty} \frac{g(L)}{L (\log L)^\eta} = \infty. \quad (1.3)$$

The repetition-function statement is proved directly. Inequality (1.1) then transfers its lower bound to p ; a lower bound for p alone would not imply the repetition-function statement.

Theorem 1.3. *Let $F(x) = (\log x)^u (\log \log x)^{-\gamma}$, where $u > 0$, $\gamma \geq 0$, and either $u < 1/2$, or $u = 1/2$ and $\gamma > 1/2$. For every $C > 0$ there is $A = A(\xi, b, C, u, \gamma) > 0$ such that, for every sufficiently large X , the interval*

$$\left[X, X \exp\{AF(X)^2 \log(2F(X))\} \right] \quad (1.4)$$

contains an integer L with both $R(L) > CLF(L)$ and $p(L) > CLF(L)$. Its upper endpoint is $X^{1+o(1)}$.

Adamczewski and Bugeaud proved $p(L)/L \rightarrow \infty$ for algebraic irrationals in integer bases [1]. Bugeaud and Evertse proved the logarithmic limsup bound with $\eta < 1/11$ [2, Theorem 2.1]. The arithmetic argument here uses their repetition method and the semistable machinery of [3, Sections 8–14]. A fixed-weight theorem alone does not permit the exact ratio $\rho = r/t$ to vary from point to point. Section 5 proves the needed finite-support variant, including a bound for its height threshold; Section 4 checks semistability for every weight in the relevant family.

The subsequent counting argument uses that theorem separately at each fixed quality. If μ dyadic prefix scales yield a single dyadic approximation scale, the largest prefix yields quality at least a constant times $2^\mu/B$. Summation over the multiplicity levels gives a bound $O(B^2 \log(2B))$ for the original number of scales. This replaces greedy thinning of the prefix scales. The proof of this step is in Section 6.

Constants may depend on ξ, b , and on the fixed window or function parameters explicitly indicated. No effective numerical starting length or assertion uniform in varying algebraic numbers or bases is made. The endpoint statement is a positive limsup; it does not assert a positive lower bound at every sufficiently large length.

2 From repetitions to approximations

Lemma 2.1. *Suppose that the prefix of length H contains two occurrences of a word of length k . Then there are integers r, s, t, v, a such that*

$$t = r + s, \quad 0 \leq r < t, \quad t + v \leq H, \quad s \geq v \geq k/3, \quad 0 \leq a \leq b^t - b^r, \quad (2.1)$$

and

$$|(b^t - b^r)\xi - a| \leq b^{-v}. \quad (2.2)$$

Proof. Let D be the difference of the two starting positions. If $D \geq k$, the two occurrences are disjoint and we take $v = k$. If $D < k$, their union is a word of length $k + D$ with period D . Its first two blocks of length $v' = D \lfloor (k + D)/(2D) \rfloor$ are equal and disjoint. Here $v' \geq k/3$: if $D \geq k/3$, use $v' \geq D$; otherwise use $v' > (k - D)/2$. Thus in either case the prefix has a decomposition $UVWVX$, with $v = |V| \geq k/3$.

Put $r = |U|$, $s = |VW|$, $t = r + s$, and $A_j = \sum_{i=1}^j d_i b^{j-i}$, $A_0 = 0$. Set $a = A_t - A_r$. The number $a/(b^t - b^r)$ has expansion $U(VW)^\infty$ and agrees with ξ through the first $t + v$ digits. Its distance from ξ is at most b^{-t-v} , which proves (2.2). If P is the integer value of VW , then $a = (b^s - 1)A_r + P$, giving $0 \leq a \leq b^t - b^r$. The closed interval of values with a specified finite prefix also covers a periodic tail consisting entirely of the digit $b - 1$. The remaining length bounds follow from the disjoint occurrences. $\square$

Lemma 2.2 (Repetitions in a prescribed window). *Let $2 \leq X < Y$, let F be positive and nondecreasing on $[X, Y]$, and write $F_Y = F(Y)$. Suppose*

$$R(k) \leq CkF(k) \quad \text{for every integer } k \in [X, Y], \quad CF_Y \geq 1.$$

For each integer i with $8CF_Y X \leq 2^i \leq Y$, there is an approximation satisfying (2.1)–(2.2), with $H = 2^i$, such that

$$v_i \geq \frac{2^i}{24CF_Y}, \quad t_i \geq X/3. \quad (2.3)$$

Proof. Take $k_i = \lfloor 2^i/(4CF_Y) \rfloor$. The lower bound on 2^i gives $k_i \geq 2X - 1 \geq X$, and $CF_Y \geq 1$ gives $k_i \leq Y$. Thus $R(k_i) \leq CF_Y k_i \leq 2^i/4$, so the prefix of length 2^i contains the required repetition. Also $k_i \geq 2^i/(8CF_Y)$. Lemma 2.1 gives $v_i \geq k_i/3 \geq 2^i/(24CF_Y)$. Since $t_i \geq v_i$, the bound $t_i \geq X/3$ follows. All uses of the assumed upper bound for R occur at lengths inside $[X, Y]$. $\square$

3 Twisted heights and the exact weights

For a number field K write M_K for its places, normalized by $\|z\|_v = |z|_p^{d_v}$ on $z \in \mathbb{Q}$ when $v \mid p$, where $d_v = [K_v : \mathbb{Q}_p]/[K : \mathbb{Q}]$ and $\sum_{v \mid p} d_v = 1$. At an infinite place represented by an embedding σ_v the corresponding formula is $\|z\|_v = |\sigma_v(z)|^{d_v}$. For a system of linearly independent forms $\mathcal{L} = (L_{iv} : v \in M_K, 1 \leq i \leq 3)$ and real weights c_{iv} , put

$$H_{\mathcal{L}, c, Q}(x) = \prod_{v \in M_K} \max_{1 \leq i \leq 3} (\|L_{iv}(x)\|_v Q^{-c_{iv}}). \quad (3.1)$$

For $x \in \overline{\mathbb{Q}}^3$ use a finite extension E/K containing its coordinates, keep the forms unchanged at $w \mid v$, and multiply the weights by $d(w \mid v) = [E_w : K_v]/[E : K]$. This gives an extension-independent height.

Choose a finite Galois extension $K/\mathbb{Q}$ in $\mathbb{C}$ containing ξ , and let $G = \text{Gal}(K/\mathbb{Q})$ and $d = [\mathbb{Q}(\xi) : \mathbb{Q}] \geq 2$. Write $\beta_1, \dots, \beta_d$ for the distinct conjugates of ξ . They are all nonzero. For each infinite place v , choose $\sigma_v \in G$ with $\|z\|_v = |\sigma_v(z)|^{d_v}$, and set $\beta_v = \sigma_v^{-1}(\xi)$. Define the fixed forms by

$$(L_{1v}, L_{2v}, L_{3v}) = \begin{cases} (X_1, X_2, L_{\beta_v}), & v \mid \infty, \\ (X_1, X_2, X_3), & v \nmid \infty, \end{cases} \quad L_\beta = -\beta X_1 + \beta X_2 + X_3. \quad (3.2)$$

Their local determinants are all 1, and there are $d + 3$ distinct forms. For every rational vector x ,

$$\|L_{\beta_v}(x)\|_v = |\sigma_v(L_{\beta_v}(x))|^{d_v} = |L_\xi(x)|^{d_v}. \quad (3.3)$$

This is the inverse-conjugate transport used in [2, Section 5, equations (5.14)–(5.17)]. It uses smallness only at the specified real value of ξ .

Lemma 3.1 (Mass of a conjugate). *For every conjugate β of ξ ,*

$$\sum_{\substack{v|\infty \\ \beta_v=\beta}} d_v = \frac{1}{d}. \quad (3.4)$$

Proof. Let $\iota \in G$ be complex conjugation on K and $H = \langle \iota \rangle$. Infinite places are represented by the right cosets $H\sigma$. Since ξ is real, $(\iota\sigma)^{-1}(\xi) = \sigma^{-1}(\xi)$, so β_v does not depend on the representative. The field K is either totally real or totally imaginary, and $d_v = |H|/|G|$ in either case. The set of $\sigma \in G$ with $\sigma^{-1}(\xi) = \beta$ has cardinality $|G|/d$ by orbit–stabilizer. It is a union of right H -cosets, each of cardinality $|H|$. Its corresponding total place weight is therefore $(|G|/(d|H|))(|H|/|G|) = 1/d$. $\square$

For each rational prime $p \mid b$, put

$$\omega_p = \frac{\text{ord}_p(b) \log p}{\log b}, \quad \kappa_v = d_v \omega_p \quad (v \mid p). \quad (3.5)$$

These numbers are positive and satisfy $\sum_{p \mid b} \omega_p = 1$ and $\sum_{v \mid b} \kappa_v = 1$, where $v \mid b$ means $v \mid p$ for some $p \mid b$. Let S consist of all infinite places and all places above the prime divisors of b .

Fix $0 < \varepsilon \leq 1/2$ and $0 \leq \rho \leq 1 - \varepsilon$. In the order of (3.2), define raw weights by

$$e_v = \begin{cases} d_v(1, \rho, -\varepsilon), & v \mid \infty, \\ \kappa_v(-1, -\rho, 0), & v \mid b, \\ (0, 0, 0), & v \notin S. \end{cases} \quad (3.6)$$

Put

$$\bar{e}_v = \frac{1}{3} \sum_i e_{iv}, \quad D_\varepsilon = 1 + \varepsilon/3, \quad c_{iv}(\rho) = \frac{e_{iv} - \bar{e}_v}{D_\varepsilon}, \quad \delta = \frac{\varepsilon}{3 + \varepsilon}. \quad (3.7)$$

The two mass identities give

$$\sum_i c_{iv} = 0, \quad \sum_v \max_i c_{iv} = 1, \quad \sum_v \bar{e}_v = -\varepsilon/3. \quad (3.8)$$

Indeed $\sum_{v,i} e_{iv} = -\varepsilon$ and the sum of the raw local maxima is 1; centering changes the latter to $1 + \varepsilon/3$.

Consider any approximation satisfying (2.1)–(2.2). Choose $0 < \varepsilon \leq \min(v/t, 1/2)$ and put $\rho = r/t$. Since $t - r \geq v$, we have $\rho \leq 1 - \varepsilon$. Let $x = (b^t, b^r, a)$, $\Psi = b^t$, and $Q = \Psi^{D_\varepsilon}$. At the infinite places, (2.2) and (3.3) give $\|L_{iv}(x)\|_v \leq \Psi^{e_{iv}(\rho)}$. At a place $v \mid p \mid b$, the same bounds follow from

$$\|b^t\|_v = \Psi^{-\kappa_v}, \quad \|b^r\|_v = \Psi^{-\kappa_v \rho}, \quad \|a\|_v \leq 1.$$

At all other finite places the coordinates are integral. Hence

$$H_{\mathcal{L}, c(\rho), Q}(x) \leq \Psi^{\sum_v \bar{e}_v} = Q^{-\delta}, \quad \delta = \frac{\varepsilon}{3 + \varepsilon}. \quad (3.9)$$

For a collection of such approximations with a common ε , the condition $t_{j+1} > 2t_j$ implies $Q_{j+1} > Q_j^2$. No primitivity of x is needed: Ψ is an external parameter, not an identification with its projective height.

4 Uniform semistability

For a q -dimensional subspace $U \subset \overline{\mathbb{Q}}^3$, define its local weight as the minimum of $\sum_{i \in I} c_{iv}$ over all q -element sets I for which $(L_{iv}|_U)_{i \in I}$ is linearly independent. Its global weight $w_c(U)$ is the sum over places. The system is semistable if $w_c(U) \leq 0$ for every nonzero proper U . Define w_e analogously. Centering gives

$$w_c(U) = \frac{w_e(U) + q\varepsilon/3}{D_\varepsilon}. \quad (4.1)$$

Proposition 4.1. *For the family (3.6)–(3.7), all nonzero proper subspaces have negative weight. More precisely,*

$$w_c(U) \leq -\frac{\varepsilon}{6D_\varepsilon} \quad (\dim U = 1), \quad w_c(U) \leq -\frac{\varepsilon}{3D_\varepsilon} \quad (\dim U = 2).$$

Proof. By Lemma 3.1, the infinite-place contribution to $w_e(U)$ can be computed using each distinct L_β once, with raw weights $(1, \rho, -\varepsilon)/d$. The finite contributions combine into one minimum calculation with coordinate forms and weights $(-1, -\rho, 0)$, since $\sum_{v|b} \kappa_v = 1$. This grouping concerns only the subspace weight calculation; all original places and norms remain in the height.

Consider a line U . If no L_β vanishes on it, its infinite contribution is $-\varepsilon$ and its finite contribution is at most zero. If two distinct L_β vanish, subtracting them shows $U = \langle (1, 1, 0) \rangle$; all L_β then vanish and $w_e(U) = \rho - 1 \leq -\varepsilon$. Otherwise precisely one L_β vanishes. If $X_1|_U \neq 0$, the exceptional infinite contribution is at most $1/d$, the remaining infinite contribution is $-(1 - 1/d)\varepsilon$, and the finite contribution is -1 . Thus $w_e(U) \leq -(1 - 1/d)(1 + \varepsilon)$. If $X_1|_U = 0$, then $X_2|_U \neq 0$ and $w_e(U) = -(1 - 1/d)(\rho + \varepsilon)$. Since $d \geq 2$, every line therefore has $w_e(U) \leq -\varepsilon/2$.

For planes write $U = \ker M$, and put $A_\beta = \text{span}(L_\beta, X_2)$ and $B = \text{span}(X_1, X_2)$ in the dual space. For distinct conjugates β, β' ,

$$A_\beta \cap A_{\beta'} = A_\beta \cap B = \langle X_2 \rangle.$$

This follows by comparing the X_1 and X_3 coefficients. At a grouped infinite place the minimum pair has weight $(\rho - \varepsilon)/d$ unless $M \in A_\beta$; the finite minimum is $-1 - \rho$ unless $M \in B$. The following list is exhaustive. Proportional normals are identified.

Position of M	$w_e(\ker M)$
$M \notin B \cup \bigcup_\beta A_\beta$	$-1 - \varepsilon$
$M \in A_\beta \setminus (\langle L_\beta \rangle \cup \langle X_2 \rangle)$	$-1 - \varepsilon + (1 - \rho)/d$
$M \in \langle L_\beta \rangle$ for some β	$-(1 - 1/d)(1 + \varepsilon)$
$M \in B \setminus (\langle X_1 \rangle \cup \langle X_2 \rangle)$	$\rho - 1 - \varepsilon$
$M \in \langle X_1 \rangle \cup \langle X_2 \rangle$	$-\varepsilon$

Each entry is at most $-\varepsilon$, since $d \geq 2$, $\varepsilon \leq 1/2$ and $\rho \leq 1 - \varepsilon$. For example, at $M = L_\beta$ the pair for that conjugate must be (X_1, X_2) , whereas at $M = X_1$ the finite pair is (X_2, X_3) . These observations also cover the ties when $\rho = 0$. Equation (4.1) proves both bounds. $\square$

5 A finite-support moving-weight bound

We prove the arithmetic input separately. In this section K , S , and $\mathcal{L}$ are fixed; the dimension is three. Assume S is finite and contains all infinite places, $\mathcal{L}$ is the coordinate system outside S ,

and the determinant at every place is 1. Write $s = |S|$. Let $v_0 \notin S$ be a fixed finite place. Weights called admissible below satisfy

$$c_{iv} = 0 \ (v \notin S), \quad \sum_i c_{iv} = 0, \quad \sum_v \max_i c_{iv} \leq 1, \quad w_c(U) \leq 0 \ (0 \neq U \subsetneq \overline{\mathbb{Q}}^3). \quad (5.1)$$

Theorem 5.1. *There are constants depending only on the fixed data with the following property. For $0 < \delta \leq 1$ there exist $\Omega(\delta) > 2$ and $C_*(\delta) > e$ with*

$$\Omega(\delta) \ll \delta^{-5}, \quad \log \log C_*(\delta) \ll \delta^{-2} \log(2/\delta), \quad (5.2)$$

such that any sequence $(c^{(h)}, Q_h)$ of admissible weights and parameters with

$$Q_1 \geq C_*(\delta), \quad Q_{h+1} > Q_h^{\Omega(\delta)}, \quad \{x \neq 0 : H_{\mathcal{L}, c^{(h)}, Q_h}(x) \leq Q_h^{-\delta}\} \neq \emptyset$$

has length $O(\delta^{-2})$.

We first isolate the pointwise part of the proof in [3]. This is needed to control the cutoff, rather than import the larger constant in their final interval theorem.

Lemma 5.2 (Pointwise exterior data). *There are constants $A_0 > 0$ and $c_0 > 0$, depending only on $K, \mathcal{L}$, such that if c is admissible, $Q \geq \exp(A_0/\delta)$ and $H_{\mathcal{L}, c, Q}(x) \leq Q^{-\delta}$ for some nonzero x , then there is $k \in \{1, 2\}$ with the following properties. Put $N = \binom{3}{k} = 3$. There are fixed exterior forms $\hat{L}_{lv}$, depending only on $k, \mathcal{L}$, and independent vectors $z_1, z_2 \in \overline{\mathbb{Q}}^3$ spanning a plane T such that*

$$H_2(T) \geq Q^{c_0 \delta}. \quad (5.3)$$

For every place w of a common finite extension E/K ,

$$\|\hat{L}_{lw}(z_j)\|_w \leq \begin{cases} Q^{d(w|v)a_{lv}}, & w \mid v \neq v_0, \\ Q^{d(w|v_0)b_l}, & w \mid v_0, \end{cases} \quad (5.4)$$

where, with $\gamma_v = \max_i c_{iv}$,

$$\sum_l a_{lv} = 0, \quad |a_{lv}| \leq 2\gamma_v, \quad a_{lv} = 0 \ (v \notin S), \quad \sum_l b_l \leq -\delta/3, \quad |b_l| \leq 3. \quad (5.5)$$

Here H_2 is the absolute Euclidean height of the exterior coordinate vector defining a subspace, as in [3, Section 6].

Proof. Let $\lambda_1 \leq \lambda_2 \leq \lambda_3$ be the successive infima of the twisted height over $\overline{\mathbb{Q}}$. The fixed-form comparison and the absolute Minkowski theorem [3, Lemma 7.1 and Proposition 9.2] give

$$C_0^{-1} Q^{-1} \leq \lambda_1, \quad 3^{-3/2} \leq \lambda_1 \lambda_2 \lambda_3 \leq 8. \quad (5.6)$$

The product bounds use the determinant assumption. As $\lambda_1 \leq Q^{-\delta}$, enlarging A_0 ensures $\lambda_3 \geq 1$. One of the two consecutive ratios is therefore at most $Q^{-\delta/2}$; fix such a k . Its infimum subspace T_k has dimension k .

For clarity we also give the height estimate behind this assertion. Choose a basis $g_1, \dots, g_k$ with twisted heights arbitrarily close to $\lambda_1, \dots, \lambda_k$, and at each place select an independent restricted k -tuple realizing its local minimum weight. Semistability and the determinant estimate imply that the product of these local determinant norms is at most $C \lambda_1 \cdots \lambda_k \leq C' Q^{-\delta/3}$. The last inequality

follows from (5.6) and the gap, since $k(3-k)/6 = 1/3$ for $k = 1, 2$. Let $R \geq 3$ bound the number of distinct original forms. Their nonzero k -fold determinants on the basis number at most $\binom{R}{k}$. The product formula, applied to these determinants, gives the lower bound $(C''H_2(T_k))^{-\binom{R}{k}-1}$ for the same product; this is also [3, Lemma 10.2]. The fixed constants can be absorbed when $\log Q \geq A_0/\delta$, giving $H_2(T_k) \geq Q^{c_0\delta}$, for example with $c_0 = 1/(12R^3)$ after increasing A_0 .

Apply Davenport's construction [3, Lemma 11.3] separately to this height and take the $(3-k)$ th exterior power as in its equations (11.10)–(11.17). The exterior forms depend only on $k, \mathcal{L}$. Away from v_0 the exponents a_{lv} are sums of $3-k$ original weights; their sums are zero and $|a_{lv}| \leq 2\gamma_v$. The first $N-1 = 2$ exterior vectors span T , whose height equals $H_2(T_k)$ by [3, Lemma 6.1].

Here is the size check at the auxiliary place, including the cutoff. Let ν_l be the products of $3-k$ distinct successive infima. Up to a permutation, the bounds defining Q^{b_l} are $3^{27}\nu_1, \dots, 3^{27}\nu_{N-1}, 3^{27}\nu_{N-1}$. Thus their product is

$$3^{27N}(\lambda_1\lambda_2\lambda_3)^{\binom{2}{2-k}} \frac{\lambda_k}{\lambda_{k+1}} \leq CQ^{-\delta/2} \leq Q^{-\delta/3}.$$

Also (5.6) bounds any product of $p = 1, 2$ distinct infima between $C^{-1}Q^{-p}$ and CQ^{3-p} . It follows that $|b_l| \leq 3$ once A_0 is sufficiently large. All absorptions above have the form $Q^{a\delta} \geq C$ with fixed $a, C > 0$, or $Q^a \geq C$. Consequently $\exp(A_0/\delta)$ suffices throughout; the final cutoff of [3, Theorem 8.1] has not been used. $\square$

5.1 The auxiliary polynomial with changing weights

Suppose m of the parameters have the same k in Lemma 5.2; label them $h = 1, \dots, m$. Their exterior forms are consequently the same. Write a_{hlv}, b_{hl} for their exterior weights and set

$$\Gamma_v = \max_{1 \leq h \leq m} \max_i c_{iv}^{(h)}. \quad (5.7)$$

Each $\Gamma_v \geq 0$, and admissibility gives

$$\sum_{v \in S} \Gamma_v \leq s. \quad (5.8)$$

All ordinary-place vanishing conditions below are imposed only at places with $\Gamma_v > 0$.

Let $r_1, \dots, r_m$ be positive integers, $D_r = \sum_h r_h$, and let $\mathcal{U}(r)$ consist of arrays $j = (j_{hl})$ of nonnegative integers with $\sum_l j_{hl} = r_h$ for each h . Put $V = |\mathcal{U}(r)|$. For $0 < \alpha < 1$ satisfying

$$m > 2\alpha^{-2} \log(2s + 2), \quad (5.9)$$

there is a nonzero multihomogeneous polynomial P over K , of block degrees r_h , with the following properties. Expand P in the exterior forms at a place v , writing its coefficients as $d_j^{(v)}$. Then

$$d_j^{(v)} = 0 \quad \text{if } \sum_{h,l} j_{hl} a_{hlv} / r_h \geq 3m\alpha\Gamma_v, \quad v \in S, \Gamma_v > 0, \quad (5.10)$$

$$d_j^{(v_0)} = 0 \quad \text{if } \sum_{h,l} j_{hl} b_{hl} / r_h \geq -m\delta/9 + 3m\alpha, \quad (5.11)$$

$$H_2(P) \leq C_K A^{D_r}. \quad (5.12)$$

Here and below $A \geq 2$ and $C_K \geq 1$ are fixed constants independent of $m, r, c^{(h)}, Q_h$.

To prove this, choose j uniformly from $\mathcal{U}(r)$. The blocks are independent, and $r_h^{-1} \sum_l j_{hl} a_{hlv}$ has mean zero and lies in $[-3\Gamma_v, 3\Gamma_v]$. Hoeffding's inequality therefore bounds the fraction satisfying the

condition in (5.10) by $e^{-m\alpha^2/2}$. For the special place the mean of the total is at most $-m\delta/9$ and the individual summands lie in $[-3, 3]$, giving the same bound. This is precisely the block-dependent form of [3, Lemma 13.3]. The total number of imposed linear equations in the V coefficients of P is less than $V/2$ by (5.9). The changes of coordinates use fixed matrices. Consequently each row of this linear system has height at most $A_1^{D_r}$ and $V \leq (eN)^{D_r}$, uniformly in which equations were selected. The number-field Siegel lemma [3, Lemma 13.1], with exponent $U/(V - U) < 1$, gives (5.12) after increasing A . This proves all three assertions, without discretizing places or weights.

For a derivative multi-index i , use divided derivatives $P_i = (\prod_{h,l} i_{hl}!)^{-1} \partial^i P$ and write

$$|i|_r = \sum_{h,l} i_{hl}/r_h.$$

If $|i|_r \leq 2m\alpha$, expansion in the exterior forms gives

$$P_i = \sum_j d_{i,j}^{(v)} \prod_{h,l} \hat{L}_{lv}(X_h)^{j_{hl}}, \quad (5.13)$$

$$d_{i,j}^{(v)} = 0 \quad \text{if} \quad \sum_{h,l} j_{hl} a_{hlv}/r_h > 12m\alpha\Gamma_v, \quad v \in S, \quad (5.14)$$

$$d_{i,j}^{(v_0)} = 0 \quad \text{if} \quad \sum_{h,l} j_{hl} b_{hl}/r_h > -m\delta/9 + 12m\alpha, \quad (5.15)$$

$$\prod_v \max_j \|d_{i,j}^{(v)}\|_v \leq C_K A_2^{D_r}. \quad (5.16)$$

Indeed a contributing original monomial has index $j + q$, where $\sum_l q_{hl} = \sum_l i_{hl}$ in every block. Its weighted shift has absolute value at most $6m\alpha\Gamma_v$, or $6m\alpha$ at v_0 . Either strict inequality above would therefore force that original coefficient to vanish by (5.10) or (5.11). The coefficient bound follows from fixed inverse coordinate matrices and binomial coefficient bounds, exactly as in [3, Lemma 13.4]; combined with (5.12) these give a fixed A_2 . At places outside $S \cup \{v_0\}$ all the weights are zero, so no vanishing condition is required.

5.2 Nonvanishing, the product formula, and the cutoff

We now prove Theorem 5.1. Choose

$$\alpha = \frac{\delta}{1000(s+1)}, \quad m = \left\lfloor 2\alpha^{-2} \log(2s+2) \right\rfloor + 1, \quad \Omega = \frac{4m^2}{\alpha}. \quad (5.17)$$

There are only two possible k , so a sequence of at least $2m - 1$ parameters has a subsequence of m parameters with the same k . Separation is preserved. Suppose such a subsequence exists and label it by $Q_1 < \dots < Q_m$.

Choose an arbitrarily large positive integer r_1 with $r_1 \log Q_1 > \alpha^{-1} \log Q_m$ and $C_K \leq 2^{r_1}$, and put

$$r_h = \left\lceil \frac{r_1 \log Q_1}{\log Q_h} \right\rceil \quad (h > 1), \quad B = Q_1^{r_1}.$$

Then

$$B \leq Q_h^{r_h} < B^{1+\alpha}, \quad r_h \leq r_1, \quad \frac{r_h}{r_{h+1}} > \frac{\Omega}{1+\alpha} \geq \frac{2m^2}{\alpha}. \quad (5.18)$$

In particular the decreasing degree ratio has the direction required by [3, Proposition 12.1, equation (12.3)]. Construct P as above. For each exterior plane T_h , use the grid

$$\mathcal{G}_h = \{u_1 z_{h1} + u_2 z_{h2} : u_j \in \mathbb{Z}, |u_j| \leq \lceil N/\alpha \rceil\}, \quad N = 3.$$

Put $E_m = (N - 1)(3m^2/\alpha)^m$ and $A_3 = 2eA$. By (5.12) and $C_K \leq 2^{r_1}$, $e^{D_r} H_2(P) \leq A_3^{D_r}$. The other hypothesis of that nonvanishing proposition is

$$H_2(T_h)^{r_h} \geq (e^{D_r} H_2(P))^{E_m}. \quad (5.19)$$

By (5.3) and (5.18) it is enough that

$$c_0 \delta \log Q_1 \geq m E_m \log A_3. \quad (5.20)$$

The proposition then supplies $x_h \in \mathcal{G}_h \setminus \{0\}$ and $|i|_r \leq 2m\alpha$ such that $P_i(x_1, \dots, x_m) \neq 0$.

We estimate this value. For an infinite place w , let $s(w)$ be its local degree divided by $[E : \mathbb{Q}]$, and set $s(w) = 0$ at finite places. The grid and (5.4) give the extra factor

$$\|\hat{L}_{lw}(x_h)\|_w \leq (N^2/\alpha)^{s(w)} Q_h^{d(w|v)a_{hlv}} \quad (w \nmid v_0), \quad (5.21)$$

with the same bound at $w \mid v_0$ using b_{hl} and no grid factor. The factor N^2/α is retained throughout.

For a surviving monomial at $w \mid v \neq v_0$, put $A_{hw} = r_h^{-1} \sum_l j_{hl} d(w \mid v) a_{hlv}$ and $\Gamma_w = d(w \mid v) \Gamma_v$. The derivative degrees are at most r_h , so $|A_{hw}| \leq 3\Gamma_w$. Replacing $Q_h^{r_h}$ by B costs at most $3m\alpha\Gamma_w$ in its exponent, by (5.18). Equation (5.14) bounds the remaining sum by $12m\alpha\Gamma_w$. The resulting exponent is at most $15m\alpha\Gamma_w$. At $w \mid v_0$ the same argument using (5.15) gives at most $d(w \mid v_0)(-m\delta/9 + 15m\alpha)$.

The number of monomials in a derivative is at most $V \leq (eN)^{D_r}$. Multiplying the local bounds, using (5.16), $\sum_w s(w) = 1$, $\sum_{w|v} d(w \mid v) = 1$, and (5.8), gives

$$\begin{aligned} \prod_w \|P_i(x_1, \dots, x_m)\|_w &\leq G(\alpha)^{D_r} (B^m)^{-\delta/9 + 15(s+1)\alpha} \\ &\leq G(\alpha)^{D_r} (B^m)^{-\delta/12}, \end{aligned} \quad (5.22)$$

where $G(\alpha) = 2A_2 e N^3 / \alpha$. We used $C_K \leq 2^{r_1} \leq 2^{D_r}$. Since $D_r \leq m r_1$, the right side is strictly less than 1 if

$$\log Q_1 \geq 24\delta^{-1} \log G(\alpha). \quad (5.23)$$

This contradicts the product formula for the nonzero value of P_i .

For definiteness, all the required conditions hold with

$$C_*(\delta) = \exp \left(\frac{A_0 + 1}{\delta} + \frac{2mE_m \log A_3}{c_0 \delta} + \frac{24 \log G(\alpha)}{\delta} \right). \quad (5.24)$$

As $m \ll \delta^{-2}$ and $\alpha \asymp \delta$, $\Omega \ll \delta^{-5}$ and

$$\log \log C_*(\delta) \ll m \log(m/\alpha) + \log(2/\delta) \ll \delta^{-2} \log(2/\delta).$$

The maximum length is at most $2m - 2$. This proves the theorem.

Corollary 5.3. *Under the same hypotheses, suppose $Q_{h+1} > Q_h^2$. The number of parameters at or above $C_*(\delta)$ is $O(\delta^{-2} \log(2/\delta))$.*

Proof. Set $q = \lceil \log_2 \Omega \rceil \ll \log(2/\delta)$ and partition the indices modulo q . In each part consecutive parameters satisfy the separation condition of Theorem 5.1. Apply its bound to each part. $\square$

6 Dyadic scales and fixed-quality layers

Fix the forms and support from Section 3, and choose a finite place v_0 above a rational prime not dividing b . The admissibility requirements in Section 5 follow from (3.8) and Proposition 4.1.

Lemma 6.1 (Counting all prefix scales). *There are constants $K_0, K_1 > 0$, depending only on ξ, b , with the following property. Let $B \geq 2$ and let I be a finite set of integers. For each $i \in I$, suppose there is an approximation satisfying (2.1)–(2.2), with $H = 2^i$, such that*

$$t_i \geq T, \quad v_i \geq 2^i/B.$$

If

$$\log T \geq K_0 B^2 \log(2B), \tag{6.1}$$

then

$$|I| \leq K_1 B^2 \log(2B). \tag{6.2}$$

Proof. Put $\tau_i = \lfloor \log_2 t_i \rfloor$ and $\mu_\tau = \#\{i \in I : \tau_i = \tau\}$. For each nonempty class retain its largest index i_τ and that index's approximation. Since $t_i \leq 2^i$, every member satisfies $i \geq \tau$. The indices are distinct, so $i_\tau \geq \tau + \mu_\tau - 1$. As $t_{i_\tau} < 2^{\tau+1}$, this yields

$$\frac{v_{i_\tau}}{t_{i_\tau}} > \frac{2^{i_\tau - \tau - 1}}{B} \geq \frac{2^{\mu_\tau - 2}}{B}. \tag{6.3}$$

For each integer $j \geq 1$ set

$$\mathcal{T}_j = \{\tau : \mu_\tau \geq j\}, \quad N_j = |\mathcal{T}_j|, \quad \varepsilon_j = \frac{2^{j-3}}{B}, \quad \delta_j = \frac{\varepsilon_j}{3 + \varepsilon_j}.$$

Only levels with $N_j > 0$ need consideration. At such a level, (6.3) and $v_i/t_i \leq 1$ imply $2^{j-2}/B < 1$, hence $0 < \varepsilon_j < 1/2$. Every retained approximation in that level has $v_i/t_i > 2\varepsilon_j$, so (3.9) applies with the common ε_j and δ_j .

Partition $\mathcal{T}_j$ according to the parity of τ . In either part consecutive values differ by at least two. Thus their retained parameters satisfy $t_{\text{next}} \geq 2^{\tau+2} > 2t_{\text{previous}}$, and consequently $Q_{\text{next}} > Q_{\text{previous}}^2$.

We verify the cutoff uniformly over all nonempty levels. We have $\delta_j \geq \delta_1 = 1/(12B + 1)$. The function $\delta^{-2} \log(2/\delta)$ decreases on $(0, 1]$; therefore (5.2) gives

$$\log \log C_*(\delta_j) \ll B^2 \log(2B).$$

On the other hand, $Q = b^{D_{\varepsilon_j} t}$ and $D_{\varepsilon_j} \geq 1$ give $\log \log Q \geq \log T + \log \log b$. Increasing K_0 in (6.1) puts every retained point above its level's cutoff. Monotonicity of C_* itself is not needed.

Apply Corollary 5.3 to both parity classes. Since $\delta_j \geq 2\varepsilon_j/7$, it follows that

$$N_j \ll \delta_j^{-2} \log(2/\delta_j) \ll B^2 4^{-j} \log(2B). \tag{6.4}$$

The constants are uniform in B, j and the approximations. Summation over multiplicity levels now gives

$$|I| = \sum_{\tau} \mu_\tau = \sum_{j \geq 1} N_j \ll B^2 \log(2B) \sum_{j \geq 1} 4^{-j} \ll B^2 \log(2B),$$

as required. $\square$

Remark 6.2. Each application of Corollary 5.3 uses a fixed quality δ_j . Changing the layer changes this common quality and the associated normalized heights. The uniform bound for the cutoffs makes this legitimate; no theorem for a chain with varying smallness exponents δ_h is used.

7 Window bounds and the endpoint

Proof of Theorem 1.1. Fix $\theta > 1$, let $Y = X^\theta$, and put $F(x) = \sqrt{\log x / \log \log x}$. Suppose that, throughout the integer lengths in $[X, Y]$,

$$R(k) \leq CkF(k), \quad (7.1)$$

where $C > 0$ is a fixed constant to be chosen sufficiently small. Let $F_Y = F(Y)$, $B = 24CF_Y$, and

$$I = \{i \in \mathbb{Z} : 8CF_Y X \leq 2^i \leq Y\}.$$

For sufficiently large X , $B \geq 2$ and $CF_Y \geq 1$. Lemma 2.2 provides the approximations needed in Lemma 6.1, with $T = X/3$. The window buffer ensures that every invoked value of $R(k)$ lies inside $[X, Y]$.

For fixed C, θ , the number of scales satisfies

$$|I| = \frac{\log(Y/X)}{\log 2} - O(\log F_Y + |\log C| + 1) = \left(\frac{\theta - 1}{\log 2} + o(1) \right) \log X. \quad (7.2)$$

Also

$$B^2 \log(2B) = (288C^2 + o(1)) \log Y = (288\theta C^2 + o(1)) \log X. \quad (7.3)$$

Choose $C > 0$ so small that

$$288\theta K_0 C^2 < \frac{1}{2}, \quad 288\theta K_1 C^2 < \frac{\theta - 1}{2 \log 2}.$$

Then $\log(X/3) \geq K_0 B^2 \log(2B)$ for sufficiently large X , and Lemma 6.1 applies. Its bound (6.2) contradicts (7.2)–(7.3). Hence some integer $L \in [X, X^\theta]$ has $R(L) > CLF(L)$. By (1.1), the same L has $p(L) \geq R(L) - L > (C/2)LF(L)$ once X is large enough. This proves the theorem with $c = C$. $\square$

Proof of Corollary 1.2. Take, for example, $\theta = 3/2$ in Theorem 1.1. The lengths provided tend to infinity with X , proving the positive limsup. Multiplying the resulting lower bounds by $(\log \log L)^{\gamma-1/2}$ proves the first assertion in (1.3); multiplying by $(\log L)^{1/2-\eta}/\sqrt{\log \log L}$ proves the second. $\square$

Proof of Theorem 1.3. For the given F , we have $F(X) \rightarrow \infty$ and

$$F(X)^2 \log(2F(X)) = o(\log X). \quad (7.4)$$

Fix $C_0 > 0$ and write

$$W = AF(X)^2 \log(2F(X)), \quad Y = Xe^W.$$

For fixed A , equation (7.4) gives $W = o(\log X)$. Thus $\log Y / \log X \rightarrow 1$. Substitution into the definition of F then yields $F(Y)/F(X) \rightarrow 1$.

Suppose $R(k) \leq C_0 k F(k)$ at all integer lengths in $[X, Y]$. Apply Lemma 2.2 with $B = 24C_0 F(Y)$ and $I = \{i : 8C_0 F(Y) X \leq 2^i \leq Y\}$. Since $W / \log F(X) \rightarrow \infty$, we have

$$|I| = \left(\frac{A}{\log 2} + o(1) \right) F(X)^2 \log(2F(X)), \quad B^2 \log(2B) = (576C_0^2 + o(1)) F(X)^2 \log(2F(X)).$$

The threshold (6.1), with $T = X/3$, holds by (7.4). Choose $A > 576K_1 C_0^2 \log 2$. For all sufficiently large X , (6.2) is then a contradiction. Thus some $L \in [X, Y]$ satisfies $R(L) > C_0 L F(L)$. To obtain the assertion for the prescribed C , apply this with $C_0 = 2C$. For large X , (1.1) gives $p(L) > CLF(L)$ at the same L , as well as $R(L) > CLF(L)$. Finally (7.4) gives $Y = X^{1+o(1)}$. $\square$

8 Scope and the remaining losses

The arithmetic proof uses finite common support to impose simultaneous coefficient vanishing with $m = O(\delta^{-2})$, and an explicit check of semistability for the full family of exact weights. In particular, the moving-weight theorem is proved in Section 5; it does not follow by simply applying the published fixed-weight theorem at each point. Nonreal conjugates and composite bases are handled by the mass and prime-weight identities in Section 3.

The dyadic argument records multiplicity across prefix scales. It does not give an arithmetic bound for all pairs of equal words within one prefix. If f_w counts the occurrences of a length- k word in a prefix of length H , then Cauchy–Schwarz gives

$$\sum_w f_w^2 \geq \frac{(H - k + 1)^2}{p(k)}.$$

The repetition lemma only needs one pair; multiplicities of arbitrary pairs are not retained by the subsequent arithmetic construction. Lemma 6.1 instead retains the multiplicity of prefix scales through the quality gain (6.3).

The exponent $1/2$ in the present estimates comes from (5.9), together with $\alpha \asymp \delta$ in (5.17). Two additional restrictions each contribute one logarithm at the final scale: the nonvanishing threshold has $\log \log C_* \ll \delta^{-2} \log(2/\delta)$, and passing from Ω -separation to ordinary doubling separation costs $q \asymp \log(2/\delta)$ in Corollary 5.3. These are separate requirements at the same order $B^2 \log(2B)$; their logarithmic costs are not multiplied in the dyadic proof. Both requirements must be addressed to remove the remaining $\sqrt{\log \log L}$ denominator by this method. This identifies limitations of the argument given here, without asserting an obstruction to all other methods.

The endpoint proof gives fixed-power windows with constants depending on their exponent. The proof of shrinking relative logarithmic windows uses (7.4); it does not establish such windows at $u = \gamma = 1/2$. No conclusion of positive entropy, disjunctivity, or normality follows from these block-complexity lower bounds.

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